天体核物理学

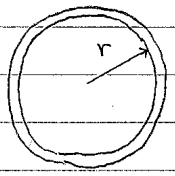
1 A.U. = 1.5×10^{13} cm

▷ Apr. 26

§2 力学平衡

球対称な星　密度 $\rho(r,t)$　$dm = 4\pi r^2 \rho(r,t)\, dr$

$Q(r,t)$



$m(r,t)$: 半径 r の内にある質量

m を独立変数にすることが多い ⇒ Lagrange coordinate

$$\frac{\partial}{\partial r} = 4\pi r^2 \rho \frac{\partial}{\partial m}$$

重力　$g = -\dfrac{Gm}{r^2}$

圧力勾配　$-\dfrac{\partial P}{\partial r} + g\rho = 0$

$$\rightarrow \boxed{\frac{\partial P}{\partial m} = \frac{Gm}{4\pi r^4}}$$

$M_\odot = 2 \times 10^{33}$ g

$R_\odot = 7 \times 10^{10}$ cm

$L_\odot = 3.9 \times 10^{33}$ erg/s

$G = 6.6 \times 10^{-8}$

星の半径 R
星の質量 M
中心の圧力 P_c
} P_c はどのくらいか (order estimate)

$$\frac{P_c}{M} \sim \frac{G \cdot \frac{M}{2}}{4\pi \left(\frac{R}{2}\right)^4}$$

$$P_c \sim \frac{GM_\odot^2}{R_\odot^4} \sim 1 \times 10^{15}\ \mathrm{dyn/cm^2}$$

平均密度 $\bar{\rho}$

$$\bar{\rho} = \frac{M_\odot}{\frac{4}{3}\pi R_\odot^3} = 1.4\ \mathrm{g/cm^3}$$

$1\,\mathrm{eV} \sim 1.6 \times 10^{-12}$ erg

$k \sim 1.38 \times 10^{-16}$ erg/

$1\,\mathrm{eV} \sim 10^4$ K

中心の温度 T_c　$P_c = \dfrac{\rho_c}{\mu m_H} k T_c$　μ 平均分子量 ~ 0.5

$$kT_c = \frac{\mu m_H}{\rho_c} P_c \sim \frac{\mu m_H}{\rho_c} \frac{2GM}{R^4} \sim \frac{\bar{\rho}}{\rho_c} \frac{\frac{4\pi}{3}R^3}{M} \cdot \frac{2GM^2}{R^4} \mu m_H \lesssim \frac{8\pi}{3} \left(\frac{GM}{R}\right) \mu m_H = 3 \times 10^{-9}\ \mathrm{erg}$$

$\left(\frac{GM}{R}\right) \sim 10^{15}$ erg/g, $\mu m_H \sim 10^{-24}$ g

$= 3\,\mathrm{keV}$

平衡でない場合も考える.

$$\frac{\partial^2 r}{\partial t^2} = -\frac{1}{\rho}\frac{\partial P}{\partial r} - \frac{Gm}{r^2} \rightarrow \frac{1}{4\pi r^2}\frac{\partial^2 r}{\partial t^2} = -\frac{\partial P}{\partial m} - \frac{Gm}{4\pi r^4}$$

自由落下 (free fall)

$$\frac{1}{4\pi R^2} \cdot \frac{R}{\tau_{ff}^2} = \frac{GM}{4\pi R^4} \rightarrow \tau_{ff} = \frac{1}{\sqrt{\frac{GM}{R^3}}} \sim \frac{1}{\sqrt{G\bar{\rho}}}$$

膨張の時間 T_{exp}

$$\frac{\partial^2 r}{\partial t^2} = -\frac{1}{\rho}\frac{\partial P}{\partial r} \qquad c_s = \sqrt{\gamma \frac{P}{\rho}}$$

$$\frac{R}{T_{exp}^2} = \frac{P}{\rho R} \qquad T_{exp} \sim \frac{R}{\sqrt{P/\rho}} \sim \frac{R}{c_s}$$

§3 ビリアル定理

Nuclear energy がない

エネルギー放出率 L (erg/s) $\qquad L + \frac{dE_{tot}}{dt} = 0$

$$L = -\frac{dE_{tot}}{dt} = -\frac{1}{2}\frac{dE_g}{dt} \geq 0$$

エネルギーが出る ⇒ 星が縮む ⇒ 温度が上がる

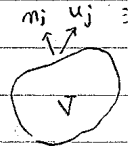
$$\frac{dE_{int}}{dt} = -\frac{1}{2}\frac{dE_g}{dt} \geq 0$$

▷ May 10

ビリアル方程式

連続の式 $\quad \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = 0$

$$\rho\left(\frac{\partial}{\partial t} + u_j \frac{\partial}{\partial x_j}\right) u_i = -\frac{\partial P}{\partial x_i} - \rho \frac{\partial \Psi}{\partial x_i}$$



$$\frac{\partial}{\partial t}\int_V \rho dV = -\int_S \rho u_j n^j dS = -\int_V \frac{\partial}{\partial x_j}(\rho u_j) dV$$

momentum $\quad \frac{\partial}{\partial t}\int_V \rho u_i dV = -\int_S \rho u_i u_j n^j dS - \int_S P n_i dS + \int_V \rho g_i dV \qquad g_i = -\frac{\partial \Psi}{\partial x_i}$

$$= -\int_V \frac{\partial}{\partial x_j}(\rho u_i u_j) dV - \int_V \frac{\partial P}{\partial x_i} dV - \int_V \rho g_i dV$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial P}{\partial x_i} - \rho \frac{\partial \Psi}{\partial x_i}$$

$$\Downarrow$$

$$\rho\left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j}\right) + u_i \underbrace{\left(\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j}\right)}_{=0}$$

$$\Psi = \int \frac{\rho(x')\, d^3x'}{|x - x'|} \qquad \triangle\Psi = 4\pi G\rho \qquad \triangle \frac{1}{|x-x'|} = -4\pi\delta(|x-x'|)$$

① そのまま積分

$$\int d^3x \;\rightarrow\; \frac{d}{dt}\int_V \rho u_i\, d^3x = 0 \;\rightarrow\; \frac{d^2}{dt^2}\underbrace{\int_V \rho x_i\, d^3x}_{M} = 0$$

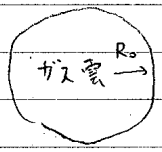
全運動量の保存

② x_j をかけた時

$$\frac{1}{2}\frac{d^2}{dt^2} I_{ij} = 2K_{ij} + B_{ij} + \delta_{ij}\Pi \qquad \text{(テンソルビリアル方程式)}$$

I_{ij}：慣性モーメント　K_{ij}：運動エネルギー　B_{ij}：重力ポテンシャル　Π：圧力

$$\frac{d^2}{dt^2}\,\mathrm{trace}\, I_{ij} = 0$$

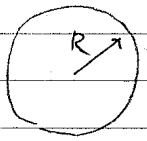


重力エネルギー $(E_g)_0 = -\dfrac{GM_0^2}{2R_0}$

$(E_{int})_0 \ll |(E_g)_0|$

重力収縮 ⟹ エネルギーを失なわなかった

平衡になって



$$E_{tot} = E_{int} + E_g = -\frac{GM^2}{2R_0}$$

E_{tot} が変化しない

↓

$R = \dfrac{R_0}{2}$ ：半分しか縮まない

$\gamma = \dfrac{5}{3}$; $E_{int} = \dfrac{1}{2}E_g$

$E_{tot} = \dfrac{1}{2}E_g = -\dfrac{GM^2}{4R}$

§4　エネルギー保存

§5　エネルギー輸送

▷ May 17

エネルギー中心で発生 → 表面で放射

Radiation

$\bar{\rho} \sim 1\,g/cc$

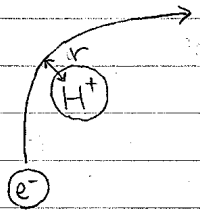
$n_e \sim 10^{24}$ 個/cc

例) 電子散乱　$\sigma_e = \dfrac{8\pi}{3} r_e^2 = 6.7 \times 10^{-25}\,cm^2$　　$\dfrac{e^2}{r_e} = m_e c^2$

$l_{ph} = \dfrac{1}{n_e \sigma_e}$　　$l < l_{ph}$ まっすぐ進める

$F = -D \nabla (aT^4)$ ← aT^4 $\quad a \sim 7 \times 10^{-15}$ erg/cm³/K⁴

$D = \frac{1}{3} l v = \frac{1}{3} l_{ph} c$



90°曲がる $\quad \frac{e^2}{r} = kT = \frac{1}{2} m_e v_e^2$

$r = \frac{e^2}{kT} \qquad \sigma = \pi r^2 = \pi \left(\frac{e^2}{kT}\right)^2$

$l = \frac{1}{n\sigma} \qquad e = 5 \times 10^{-10}$ esu

$kT = 10^5 K = 10 eV = 10^{-11}$ erg $\qquad \sigma = \pi \left(\frac{2.5 \times 10^{-20}}{10^{-11}}\right)^2 = 10^{-15}$ cm²

$n_e = 10^{24}$/cc $\qquad l = 10^{-9}$ cm

$l_{ph} = \frac{1}{n\sigma} = \frac{1}{(\frac{\sigma}{m_H})\rho} = \frac{1}{k\rho}$ $\qquad k$: opacity cm²/g

$F = -\frac{1}{3} \frac{c}{k\rho} \nabla aT^4 = -\frac{4ac}{3k\rho} T^3 \frac{\partial T}{\partial r}$

$4\pi r^2 F = L$

$\Rightarrow \boxed{\frac{\partial T}{\partial r} = -\frac{3k\rho}{4ac} \frac{1}{T^3} \frac{L}{4\pi r^2}}$

$\frac{\partial T}{\partial r} \sim 10^{-4}$ K/cm

外 · T | | T + 10⁻⁴ K 中心 $\qquad c a ((T+10^{-4})^4 - T^4) = c a T^4 \frac{4\Delta T}{T}$

$= 10^{10} \cdot 10^{-14} \cdot 10^{2\rho} 10^{-11}$

$l_{ph} = 1$ cm

$T \sim 10^7$ K $\qquad = 10^{13}$ erg/cm²/s

表面での flux

$= \frac{L_\odot}{4\pi R_\odot^2} = 10^{11}$ erg/cm²/s

$L = -D_m \frac{\partial T}{\partial m}$

$\varepsilon - \frac{\partial L}{\partial m} = \frac{\partial u}{\partial t} + P \frac{\partial v}{\partial t}$

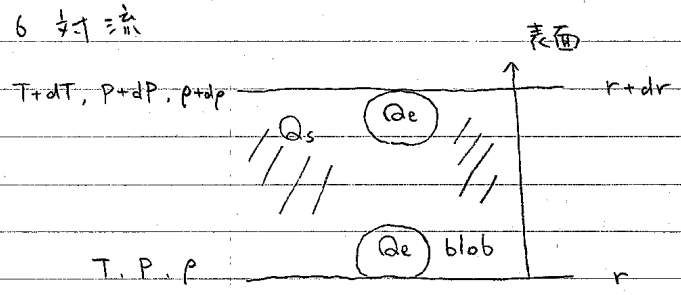
$\Rightarrow \frac{\partial}{\partial m} D_m \frac{\partial T}{\partial m} - C_v \frac{\partial T}{\partial t} = -\left[(\varepsilon - \varepsilon_\nu) + \frac{P}{\rho^2} \frac{\partial \rho}{\partial t}\right]$

$$c_v \frac{\partial T}{\partial t} = \frac{\partial}{\partial m} D_m \frac{\partial T}{\partial m}$$

$$\frac{c_v}{t_{adj}} = \frac{D_m}{M^2}$$

$$t_{adj} = \frac{c_v M}{D_m} M = \frac{c_v \bar{T} M}{L} = \frac{E_i}{L} \approx \frac{|E_g|}{L} = \tau_{KH}$$

§6 対流



$$DQ = Q_e - Q_s$$

少し温度の高い blob $\quad DT = T_e - T_s > 0 \quad T_{exp} = \frac{d}{c_s}$ で つりあう

$$DP = 0$$

$$P = \frac{k\rho T}{\mu m_H} \text{ より } \quad D\rho < 0 \quad \rightarrow \text{浮力}$$

$$D\rho|_{r+dr} = \left[\left(\frac{d\rho}{dr}\right)_e - \left(\frac{d\rho}{dr}\right)_s\right] \Delta r$$

$$k_r = -g(D\rho)$$

$D\rho < 0$ 不安定

$D\rho > 0$ 安定

安定

$$\left(\frac{\partial \rho}{\partial r}\right)_e - \left(\frac{\partial \rho}{\partial r}\right)_s > 0$$

$$\rho = \rho(P, T, \mu) \qquad d\ln\rho = \alpha\, d(\ln P) - \delta\, d(\ln T) + \varphi\, d(\ln\mu)$$

$$(d\ln\mu)_e = 0 \rightarrow \left(\alpha \frac{d\ln P}{dr}\right)_e - \left(\delta \frac{d\ln T}{dr}\right)_e - \left(\alpha \frac{d\ln P}{dr}\right)_s + \left(\delta \frac{d\ln T}{dr}\right)_s - \left(\varphi \frac{d\ln\mu}{dr}\right)_s > 0$$

$\left(\frac{d\ln T}{d\ln P}\right)$ ⇒ 温度勾配

$$\frac{\frac{1}{T}\frac{dT}{dr}}{\frac{1}{P}\frac{dP}{dr}} = \frac{P}{T}\frac{r^2}{4\pi p}\frac{3k\rho}{4ac}\frac{1}{T^3}\frac{L}{4\pi r^2} = \frac{3k\rho L}{16\pi acGmT^4} \equiv \nabla_{rad}$$

▷ May 24

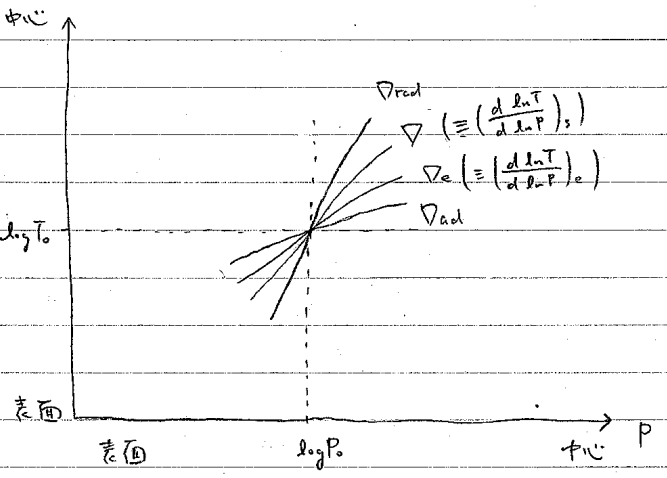
対流

Ledoux の条件

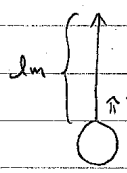
$$\nabla_{rad} < \nabla_{ad} + \frac{\varphi}{\delta}\nabla_\mu \qquad \nabla_\mu = \left(\frac{d\ln\mu}{d\ln P}\right)_S = 0 \rightarrow \text{Schwarzschild の条件}$$

Convective に不安定な時

$\nabla > \nabla_e$



Mixing length



$\xi = l_m / H_p$ レコラメ-タ-

$$(5\text{-}2)\quad F_{rad} = -\frac{4acT^3}{3k\rho}\frac{dT}{dP}\left(\frac{dP}{dr}\right) = \frac{4acT^3}{3k\rho}\frac{T}{P}\left(\frac{P}{T}\frac{dT}{dP}\right)\frac{Gm}{r^2}\rho$$

$\left(\frac{dP}{dr}\right) \Rightarrow -\frac{Gm}{r^2}\rho$ 　　$\left(\frac{P}{T}\frac{dT}{dP}\right) \Rightarrow \nabla$

$F_{conv} = \rho C_p \Delta T \cdot v$ 　　v: blob の速度

$$\frac{\Delta T}{T} = \frac{1}{T}\frac{\partial}{\partial r}(\Delta T)\frac{l_m}{2}$$

$$= \left(\left(\frac{\partial\ln T}{\partial\ln P}\right)_e - \left(\frac{\partial\ln T}{\partial\ln P}\right)_S\right)\left(\frac{\partial\ln P}{\partial r}\right)\frac{l_m}{2} = (\nabla - \nabla_e)\frac{l_m}{2H_p}$$

$\left(\frac{\partial\ln P}{\partial r}\right) \Rightarrow -\frac{1}{H_p}$

浮力 $k_r = -g(\Delta\rho/\rho)$ 単位質量当り

$$\frac{1}{2}k_r \cdot \frac{l_m}{2} = g\delta(\nabla - \nabla_e)\frac{l_m^2}{8H_p} = v^2$$

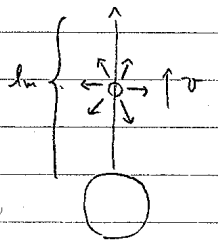
$$F_{conv} = \rho C_p T \frac{l_m^2}{4\sqrt{2}} \sqrt{g\delta} H_p^{-\frac{3}{2}} (\nabla - \nabla_e)^{\frac{3}{2}}$$

$$v = \sqrt{g\delta(\nabla - \nabla_e)\frac{l_m^2}{8H_p}}$$

d.S.V

$$\rho V C_p v \left(\frac{dT}{dr}\right)_{ad} - \lambda = \rho V C_p v \left(\frac{dT}{dr}\right)_e$$

$$\lambda = S \frac{4acT^3}{3K\rho}\left(\frac{DT}{d/2}\right) = \frac{8acT^4}{3K\rho}(\nabla - \nabla_e)\frac{S}{d}\frac{l_m}{2H_p}$$



$d = l_m$ の球だと

$$\frac{\nabla_e - \nabla_{ad}}{\nabla - \nabla_e} = \left(\frac{S l_m}{Vd}\right)\left(\frac{4acT^3}{3K\rho^2 C_p v}\right)$$

$$\frac{\delta}{l_m}^{9/2}$$

▷ May 31

§8 基礎方程式

m, t 独立変数

r, ρ, P, T

L(m), X_i

boundary

$\frac{\partial r}{\partial m}$ 中心

星の準静的
- $\rho = \rho(P, T, X_i)$
- $\frac{dP}{dm} = \cdots$ 表面
- $\frac{dL}{dm} = \cdots$ 中心
- $\frac{\partial T}{\partial m} = \cdots$ 表面
- $\frac{\partial X_i}{\partial t} = \cdots$

nonlinear

$\tau_{ff} \ll \tau_{KH} \ll \tau_n$

$t > 0$; $0 \le m \le M$

初期条件は ← X, Y, Z

M, X_i ゼロ次元 - l_m/H_p

時々刻々の平衡解は 4つの方程式

4つの条件

(1) 中心 $m = 0$

$r = 0$, $L = 0$ (P_c, T_c 不明)

(2) 表面

$P = 0$, $T = 0$ 第0近似 ← よくない

Optical depth $\tau = \int_R^\infty \sigma n dr = \int_R^\infty k\rho dr \simeq \bar{k}\int_R^\infty \rho dr = \frac{2}{3}$

$$P_{r=R} = \int_R^\infty g\rho\, dr = g_* \int_R^\infty \rho\, dr = \frac{GM}{R^2}\,\frac{2}{3}\,\frac{1}{\bar{k}}$$

$$T_{eff} = T_{r=R}$$

$$L = 4\pi R^2 \sigma T_{eff}^4$$

} boundary